

Miscellaneous Infectious Diseases of Rabbits

AKA
FOUR WAYS TO KILL THE EASTER
BUNNY



Recommend preventative medicine measures to reduce incidence of RHDV, myxomatosis, rabies, and tularemia.

Explain the main clinical presentations you would expect in rabbits infected with RHDV, myxomatosis, rabies, and tularemia.

LECTURE OUTCOMES

SAVE THE
EASTER
BUNNY



Rabbit Hemorrhagic Disease Virus

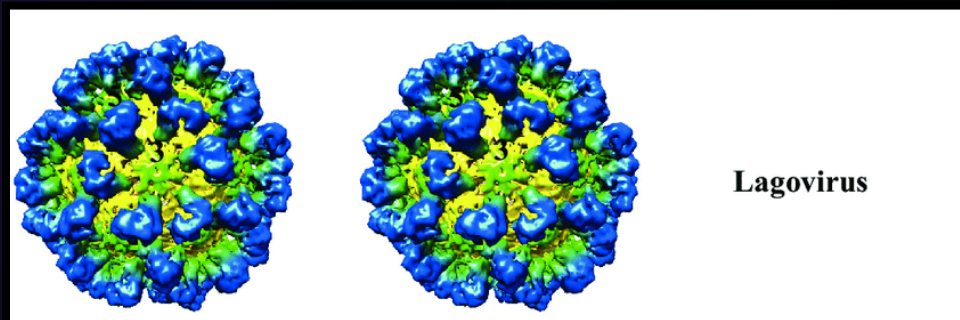
THE OTHER VIRUS OF 2020



RHDV₂

ETIOLOGIC AGENT

Highly contagious calicivirus



Nonenveloped RNA virus

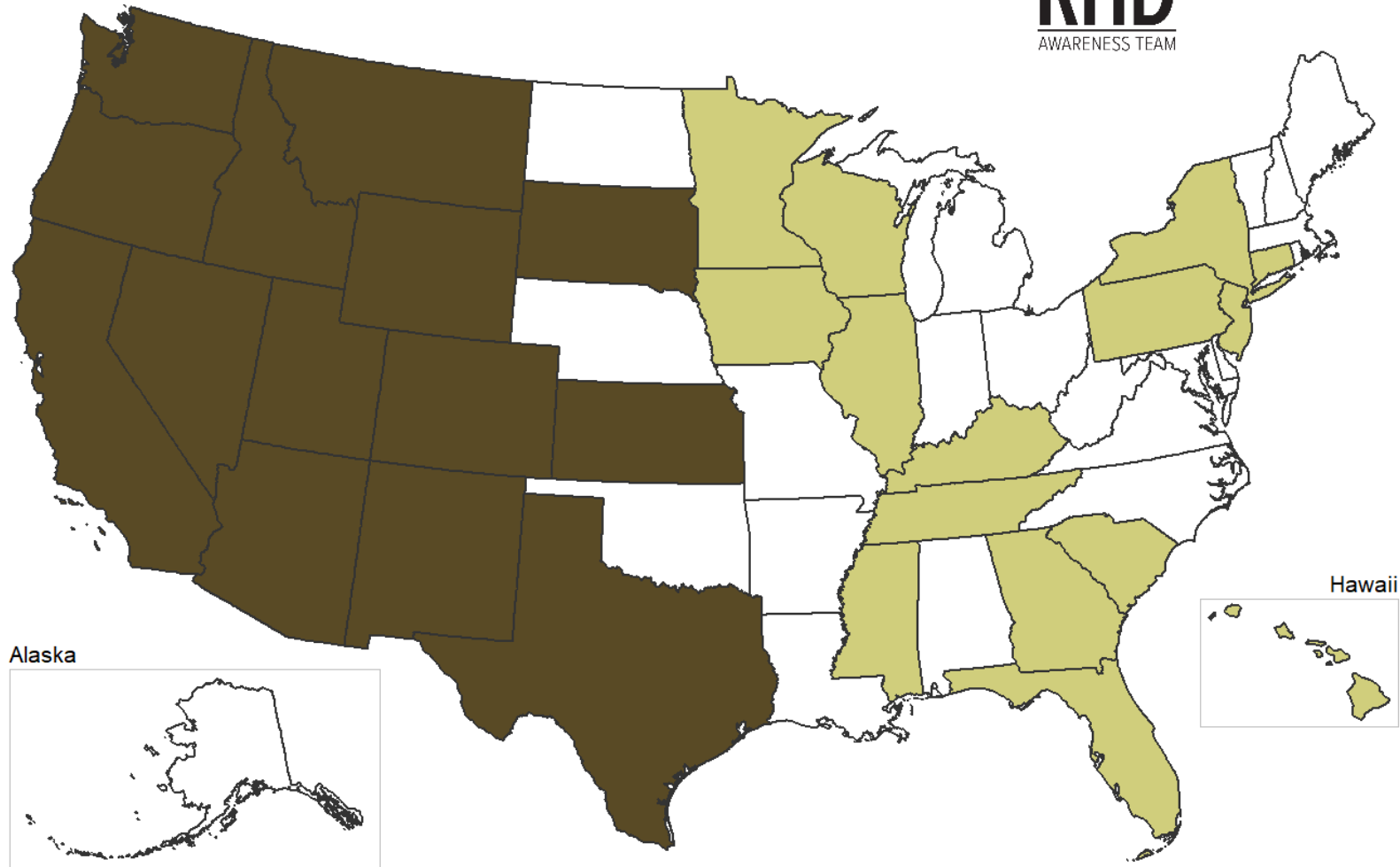
RHDV₂

HISTORY



RHDV2

January 2024



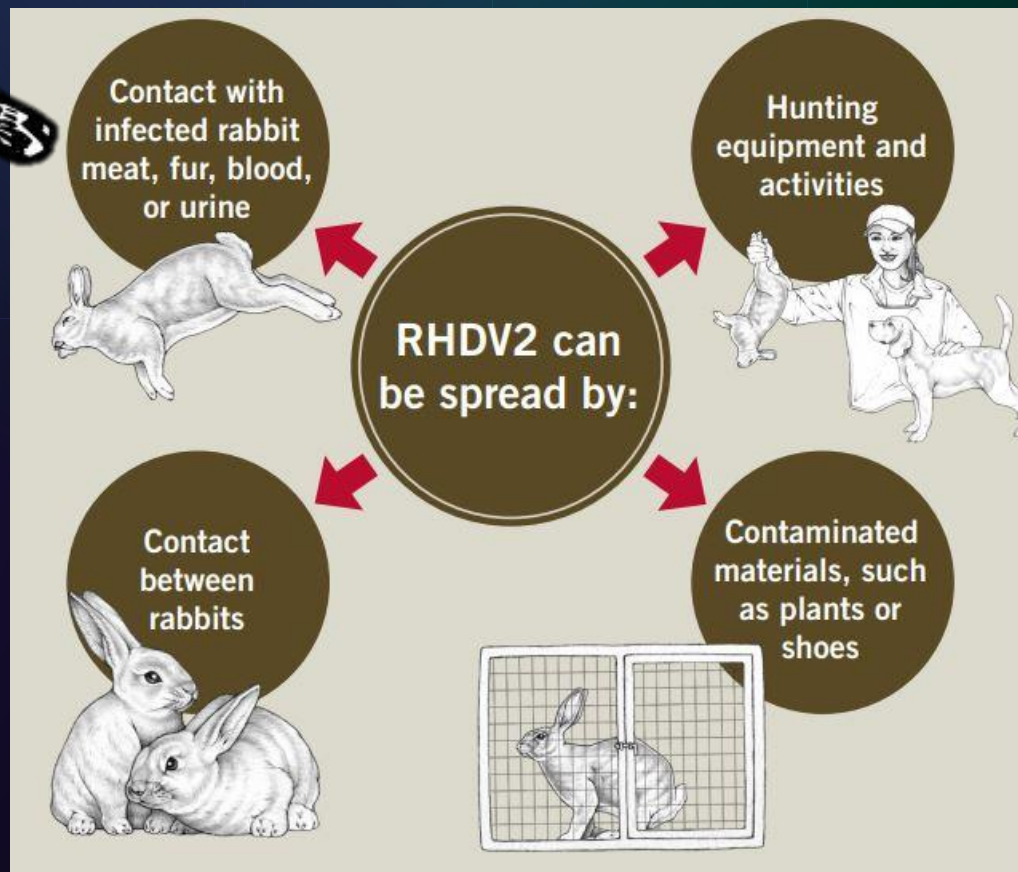
■ Domestic Cases ■ Wild Cases ■ Domestic & Wild Cases

Data provided by USDA APHIS. Supplemental data reports (e.g., Press Releases) have been collated by Rabbitats Rescue Society and WildRescue Inc. since the last USDA reporting period (every 3 months).



RHDV₂

TRANSMISSION



RHDV2.org



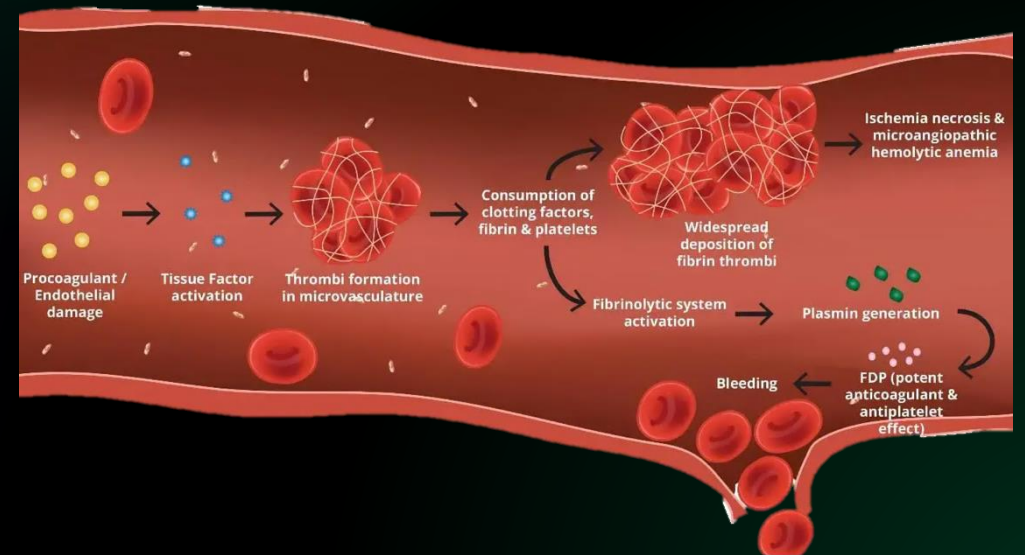
RHDV₂

PATHOPHYSIOLOGY

Target organ of infection = liver

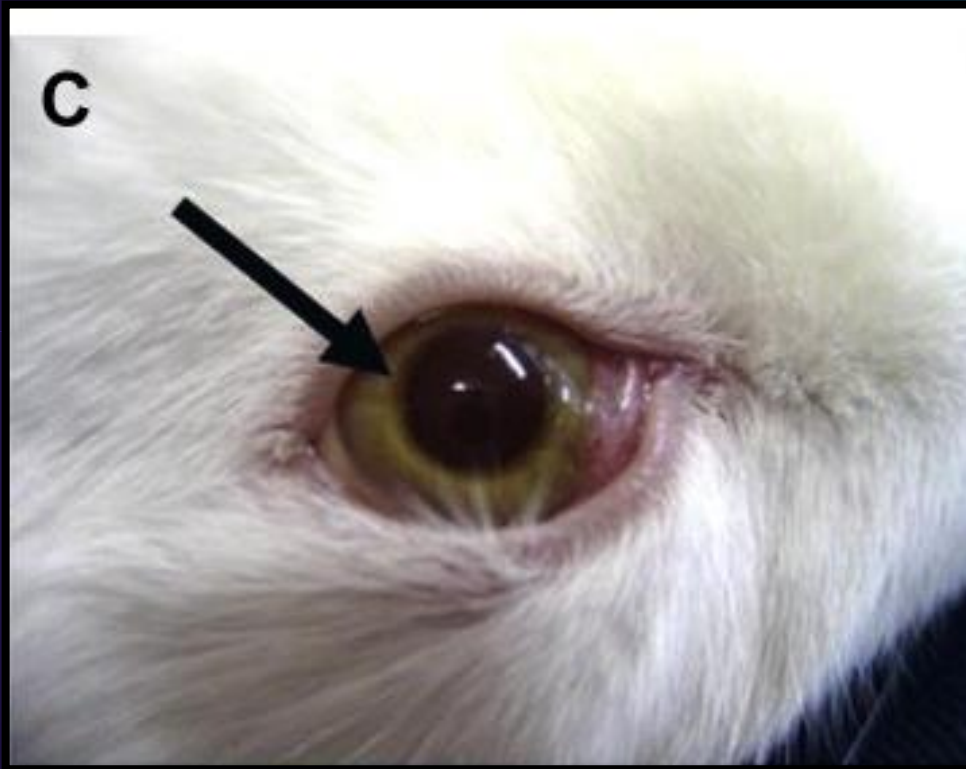
Secondary DIC and clinical
hemorrhage

Acute hepatic
necrosis



RHDV₂

CLINICAL SIGNS



<https://doi.org/10.1016/j.cvex.2013.02.002>

RHDV₂

DIAGNOSTICS



Thrombocytopenia, leukopenia

↑ Liver enzymes, ↑ bilirubin / bile acids

↑ PT, PTT, D-dimers

Confirmatory tests NOT available to practitioners

RHDV₂

THERAPEUTICS???

>80% mortality rate...





RHDV2

PREVENTION



RHDV2

PREVENTION

PROTECTING YOUR RABBIT FROM RHDV2

Rabbit Hemorrhagic Disease Virus Type-2 (RHDV2) is a highly contagious and deadly disease. The best way to protect your rabbit is to educate yourself about RHDV2, take biosecurity measures at home, and ask your veterinarian about vaccination.

- House your rabbit indoors, and reduce or eliminate outdoor playtime, especially in endemic areas.
- Wash your hands thoroughly before handling your rabbits. Change your clothes and wash your hands after contact with rabbits.
- Don't let your rabbit play with other rabbits from outside your home.
- Adopt a "no shoes" policy in your home. Disinfect shoes and other objects that may be contaminated, using a disinfectant effective for RHDV.
- Avoid hay and feed from endemic areas. Don't feed plants, grasses, or tree branches foraged from outside, especially in endemic areas.
- Trim your rabbit's nails and groom them at home. Use a monthly flea treatment (e.g., Revolution).
- Minimize your bunnies' exposure to insects such as mosquitos and flies. Maintain your window and door screens and seals.
- Quarantine any new rabbits for at least 14 days.
- Talk with your veterinarian about vaccination, in states the vaccine is available.



Learn more at
rabbit.org/rhdv



Myxomatosis

ANOTHER SPECIAL THING
CALIFORNIA HAS TO OFFER

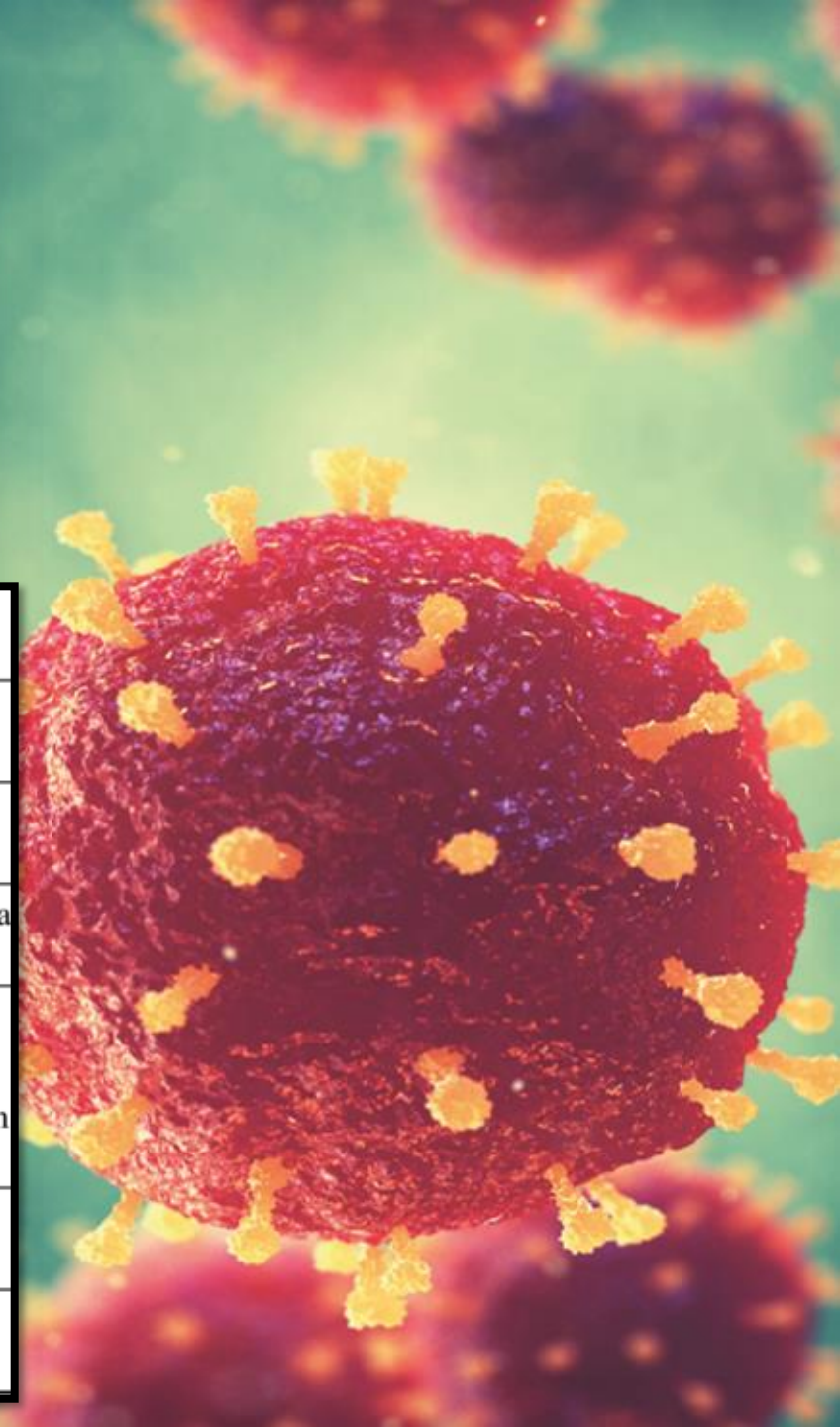


Myxomatosis

ETIOLOGIC AGENT: MYXOMA VIRUS

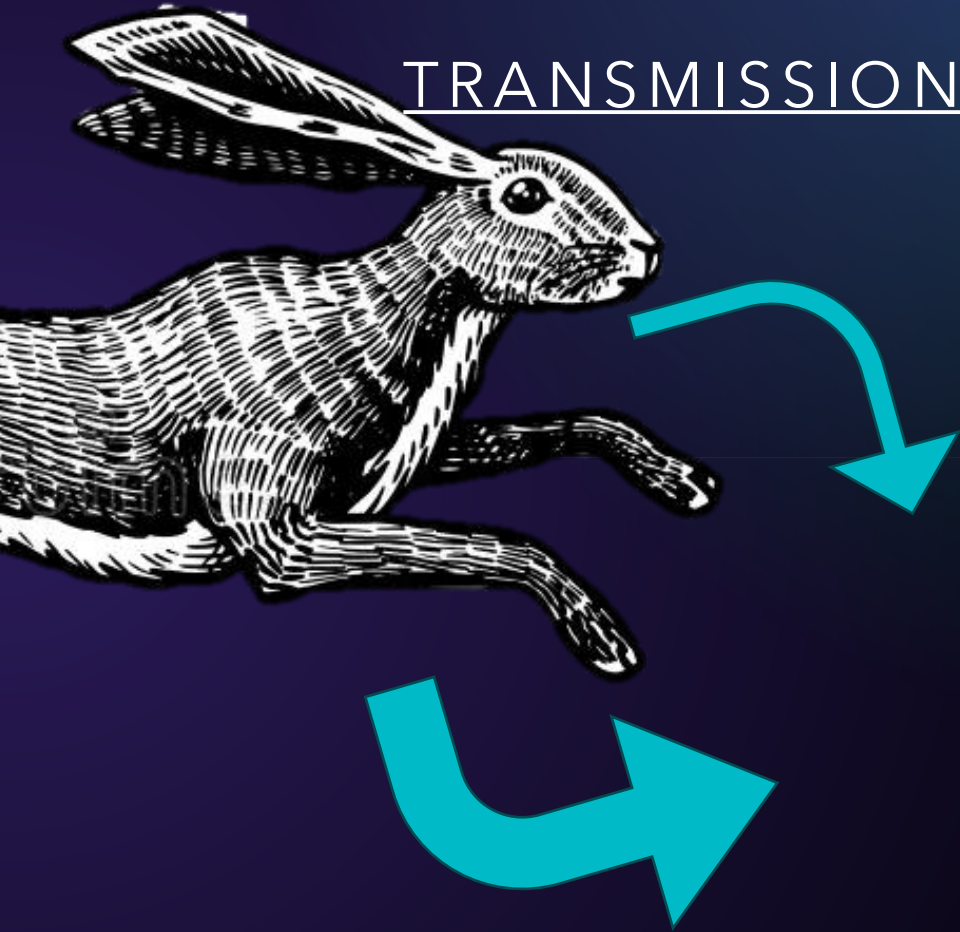
(poxvirus)

Virus	Natural Host	Disease Caused in Natural Host	Geographic Distribution of Natural Host
Myxoma virus	<i>Sylvilagus brasiliensis</i> (tapeti)	Cutaneous fibroma	South & Central America
Californian Myxoma virus	<i>Sylvilagus bachmani</i> (brush rabbit)	Cutaneous fibroma	West coast USA & Mexico
Rabbit (Shope) fibroma virus	<i>Sylvilagus floridanus</i> (eastern cottontail)	Cutaneous fibroma	Eastern & central North America to Central and South America
Squirrel fibroma virus	<i>Sciurus carolinensis</i> (eastern gray squirrel) ¹	Cutaneous epitheliofibroma; may be generalized and involve internal organs	Eastern North America; introduced in Europe and Britain
Hare fibroma virus	<i>Lepus europaeus</i> (European brown hare) ²	Cutaneous fibroma	Europe but widely introduced to other countries
Western squirrel fibroma virus	<i>Sciurus griseus griseus</i> (western gray squirrel)	Cutaneous thickening	West coast of North America

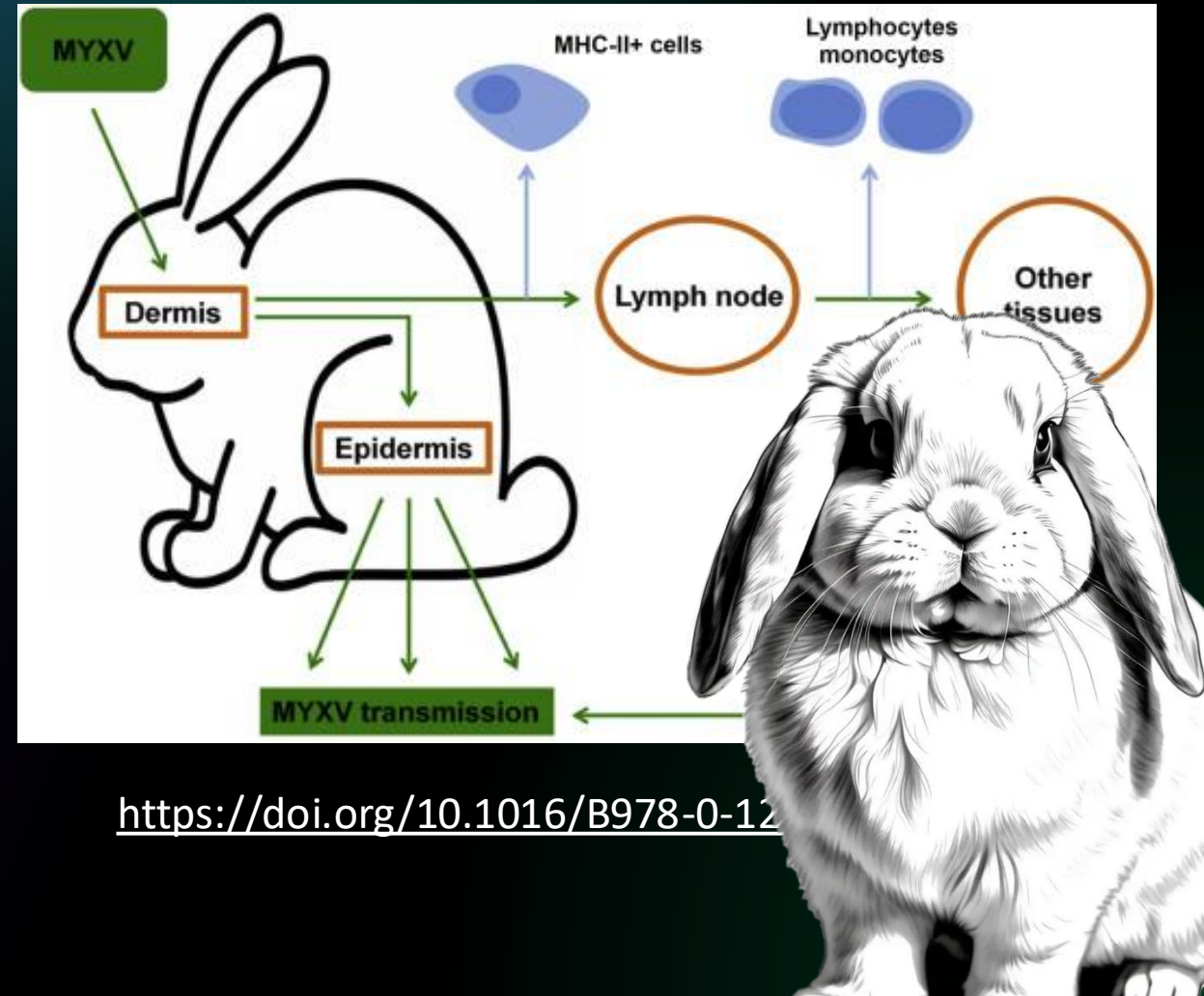


Myxomatosis

TRANSMISSION



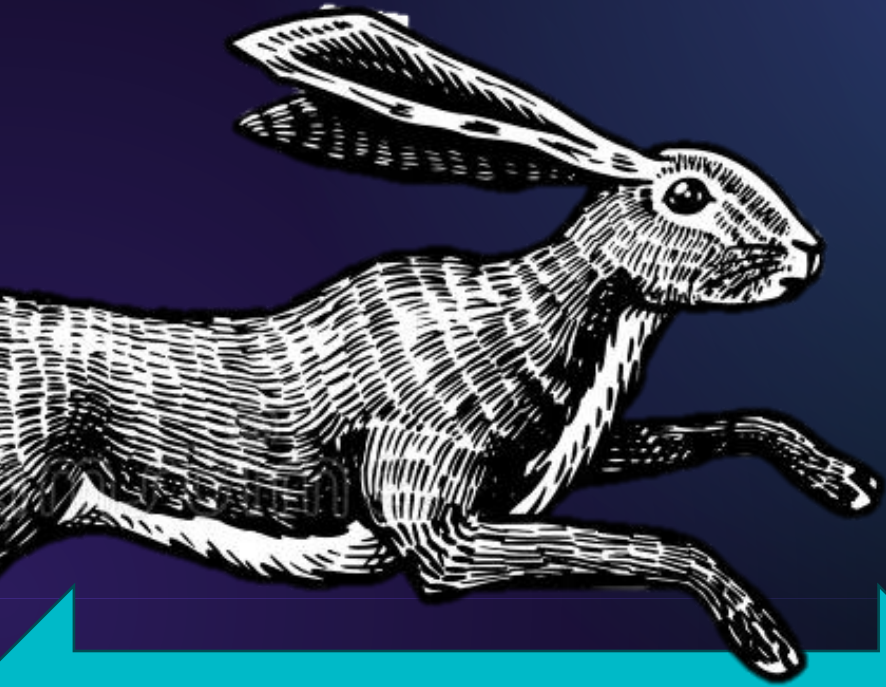
PATHOPHYSIOLOGY



<https://doi.org/10.1016/B978-0-12-811281-2>

Myxomatosis

CLINICAL SIGNS



Subclinical

Skin nodules



Facial swelling - Fever - Lymphadenopathy
DEATH

Myxomatosis

CLINICAL SIGNS



Secondary Bacterial
Infections

Myxomatosis

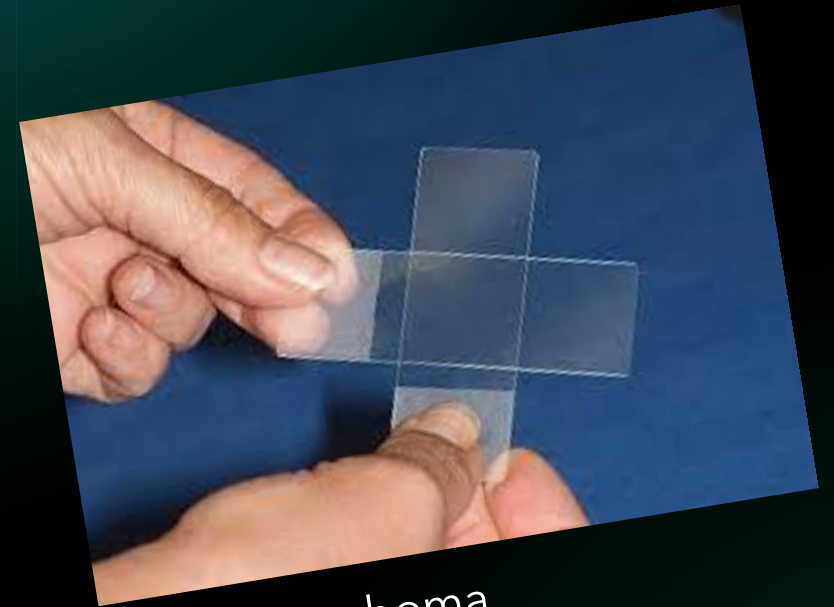
DIAGNOSTICS



Nonspecific
inflammation

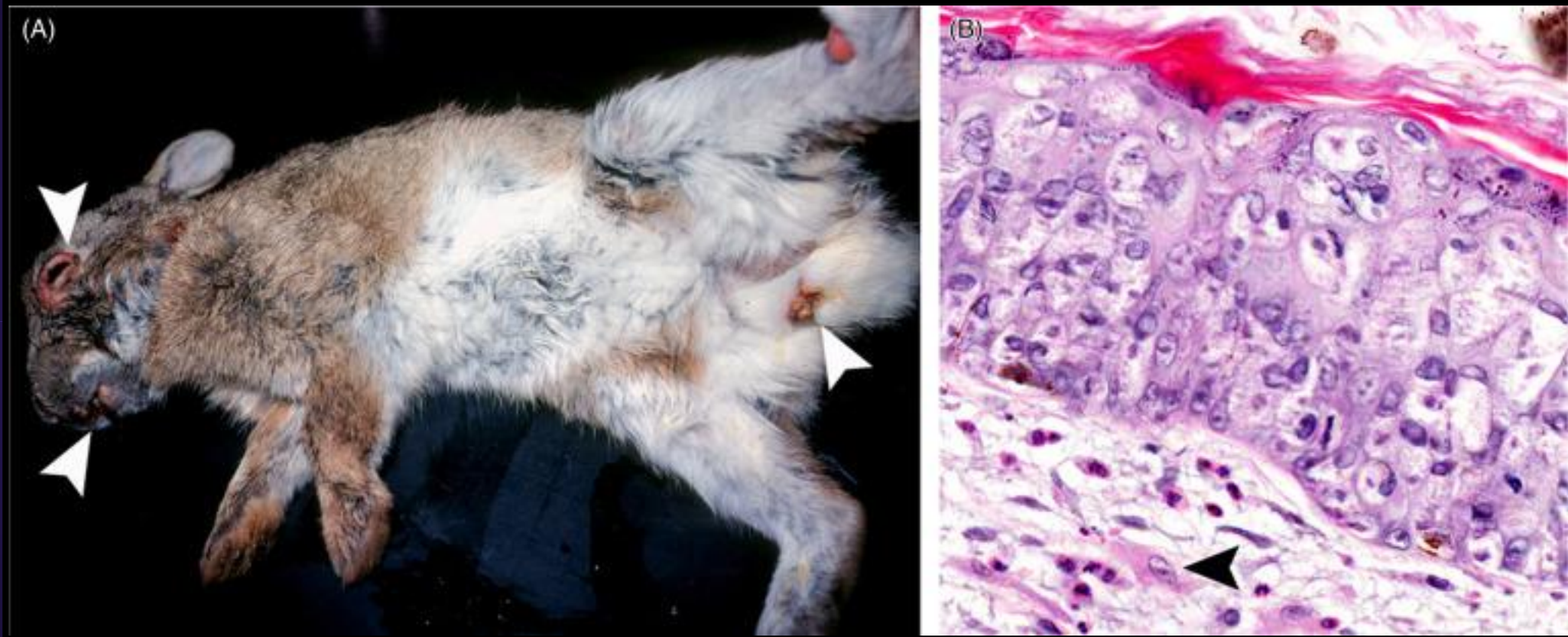


Nonspecific
findings



#notlymphoma
Reactive lymphoid tissue

Myxomatosis



Pathology of Wildlife and Zoo Animals. 2018.

Myxomatosis



THERAPEUTICS

Up to 100% mortality rate for
California myxomatosis...

Myxomatosis

PREVENTION





Tularemia

ITS ONLY FUNNY ON GRAPHIC T
SHIRTS...



Tularemia

ETIOLOGIC AGENT

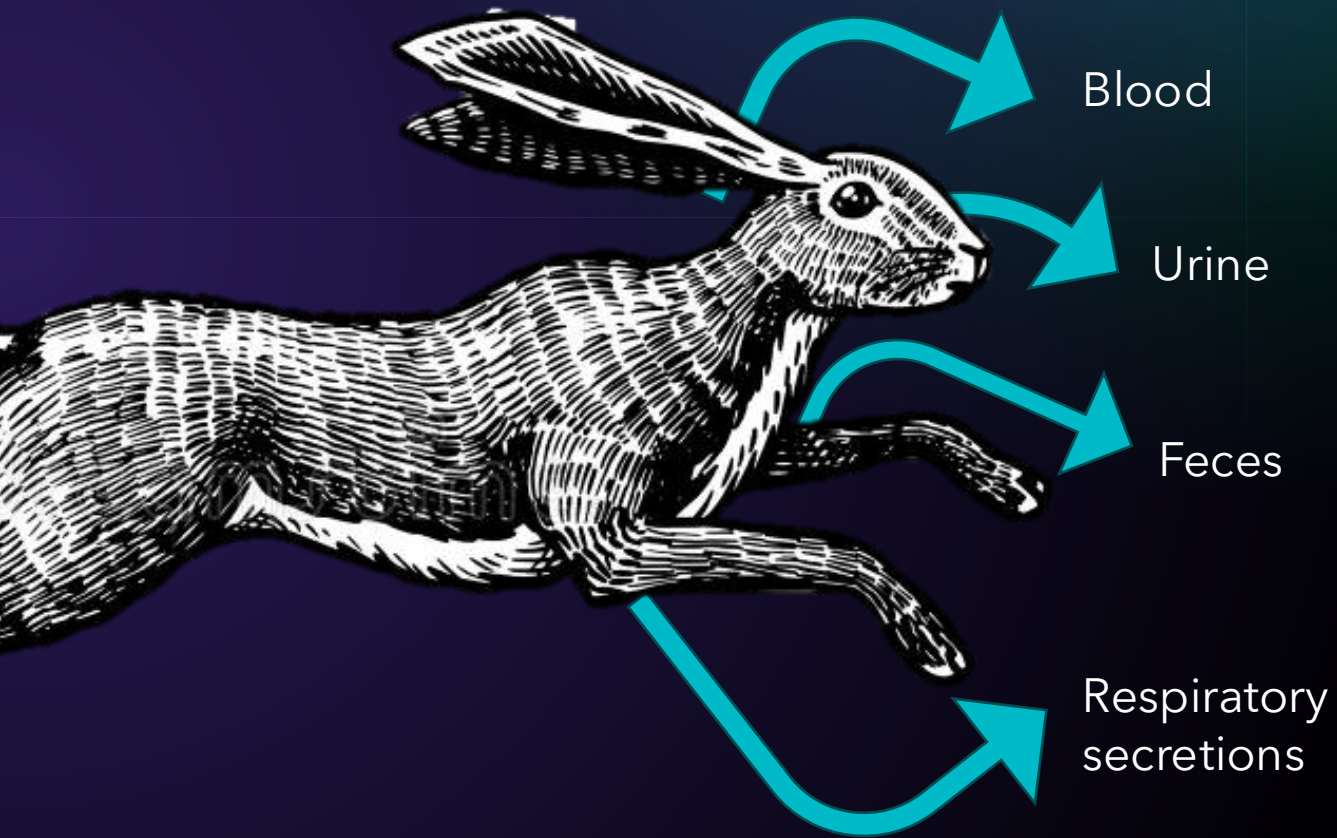
Francisella tularensis
Gram negative bacteria



Not so yummy bacteria on yummy chocolate agar...

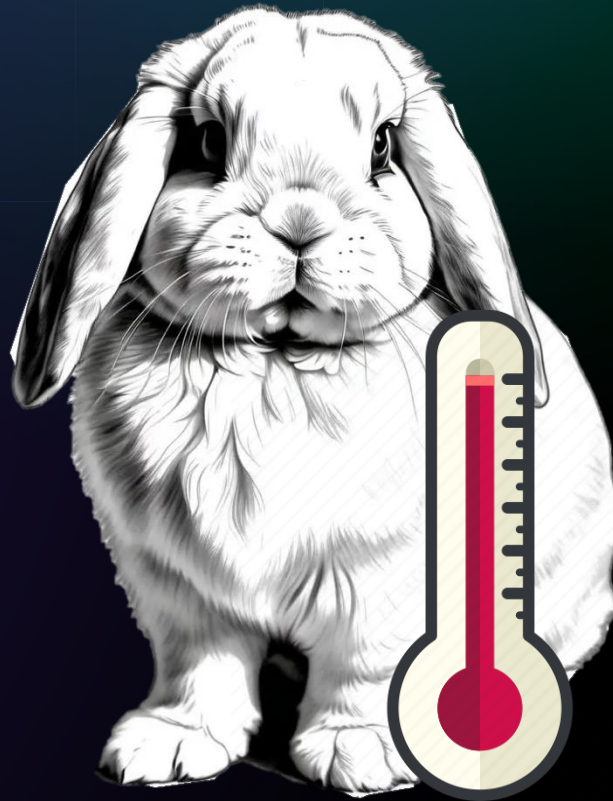
Tularemia

TRANSMISSION



Tularemia

CLINICAL SIGNS - NONSPECIFIC



Tularemia

DIAGNOSIS

Culture!



Tularemia



THERAPEUTICS

VERY high mortality rate...



Rabies

WHEN HALLOWEEN AND EASTER
COLLIDE



Rabies

ETIOLOGIC AGENT

Rabies virus – Lyssavirus

TRANSMISSION

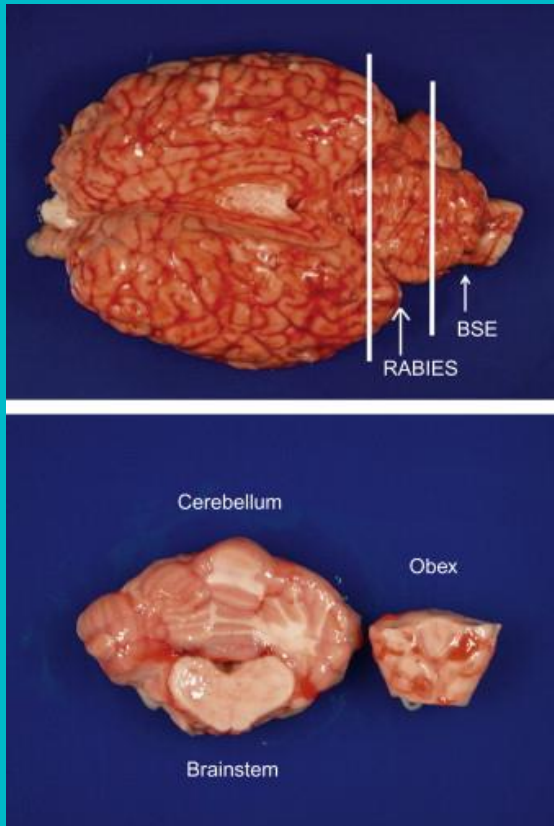
CLINICAL SIGNS

Paralytic/dumb form
Sudden death

Rabies

DIAGNOSIS

If you suspect it....



You may NOT suspect it...

- Nonspecific signs
- More likely neuro differentials
- "Has your rabbit been exposed to raccoon in the past few days?"

Rabies



THERAPEUTICS

100% mortality rate...



All of these potentially (most likely) lethal infectious diseases have outdoor access as a risk factor...

INDOOR LIVING IS BEST FOR BUNNIES



Inside I am safe, protected,
sheltered, and monitored.
I'm part of the family!



+ Housing

Rabbits should be kept as indoor pets.

3. A pet rabbit should not live outside. You will be able to enjoy & care for them better when they are inside with you. Consider these reasons.

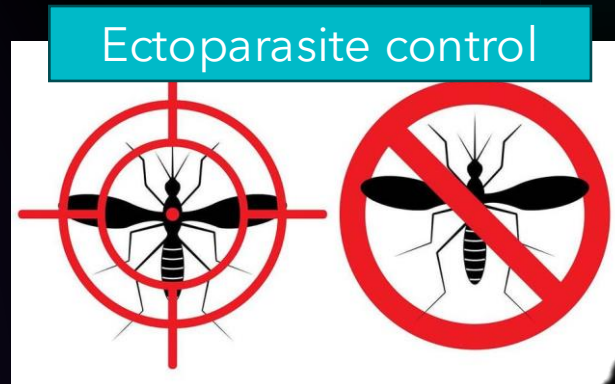


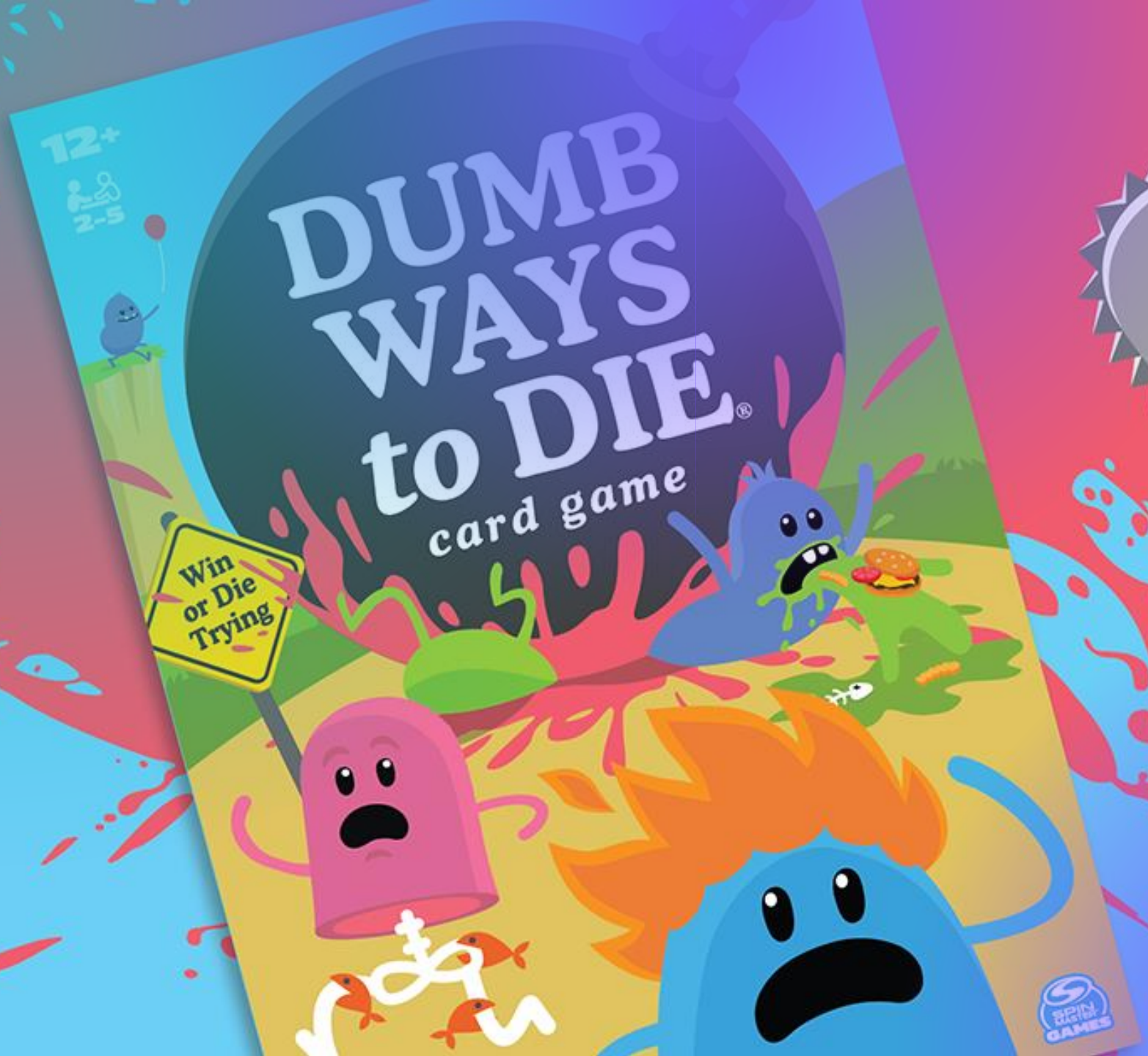


How do you counsel risk mitigation for outdoor rabbits?



Raised hutch





Hoppy
Easter!

